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Part - Desirable Judgement

(1) What is inferential statistics?

→ Inferential statistics is the process of using sample data to make conclusions or predictions about a larger population.

(2) What is Hypothesis Testing and its components?

→ Hypothesis testing is a statistical method used to decide whether a statement about a population is true or not.

Main components: Null Hypothesis (H_0), Alternative Hypothesis (H_1), Significance Level (α), Test statistic, p-value / critical value, and ~~Decision~~ Decision.

(3) Explain confidence interval and critical value?

→ confidence interval: A range of values used to estimate a population parameter with a certain level of confidence.
critical value: A boundary value used to decide whether to

reject or not reject the null hypothesis!

(4) Define p-value.

→ p-value is the probability of getting the observed result, assuming that the null hypothesis is true.
if $P\text{-value} \leq \alpha$: Reject H_0 .
if $P\text{-value} > \alpha$: Do not reject H_0 .

(5) Differentiate Type I and Type II Errors

Type I Error: Rejecting a true H_0 (False positive).

Type II Error: Not rejecting a False H_0 (False Negative).

(6) Brief Descriptions of z-test, t-test, chi-square test, and ANOVA test.

→ Z-test used to test a population mean when the sample is large or population standard deviation is known.

T-test: used to compare means when the sample is small and population standard deviation is unknown.

chi-square test: used to test the relationship between categorical variable or goodness of fit.

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 ANOVA:- used to compare the means of three or more groups.

(7) What is covariance?

→ covariance measures how two variables change together.

positive covariance: Both variables tend to increase together.

Negative covariance: one increases while the other decreases.

Zero covariance: No linear relationship.

(8) What is correlation?

→ correlation measures the strength and direction of the relationship between two variables. Its value ranges from -1 to +1.

+1: perfect positive correlation.

0: No correlation.

-1: perfect negative correlation.